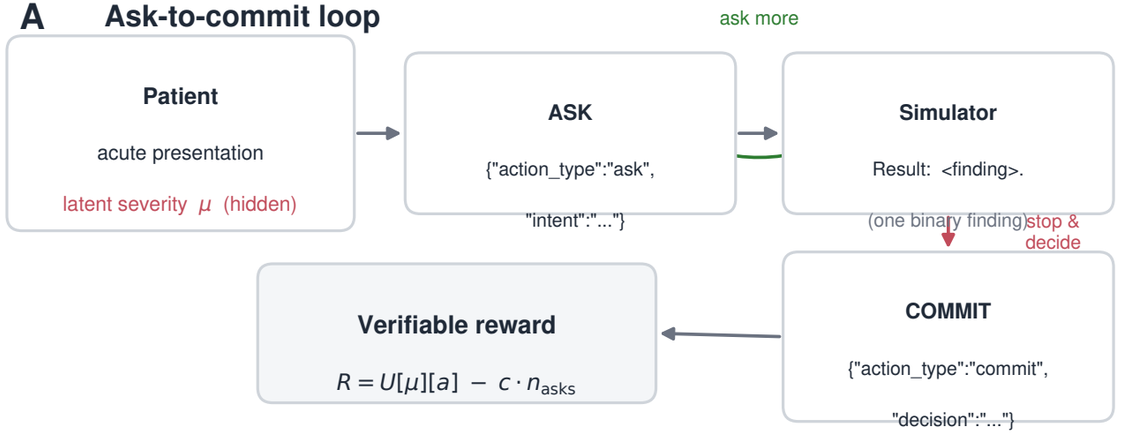
Reward Design Governs Information Gathering in Clinical Multi-Turn Agents

Clinical decisions are an ask-to-commit process; the reward geometry — not just the verifiable final score — decides whether RL learns a safe gathering procedure.

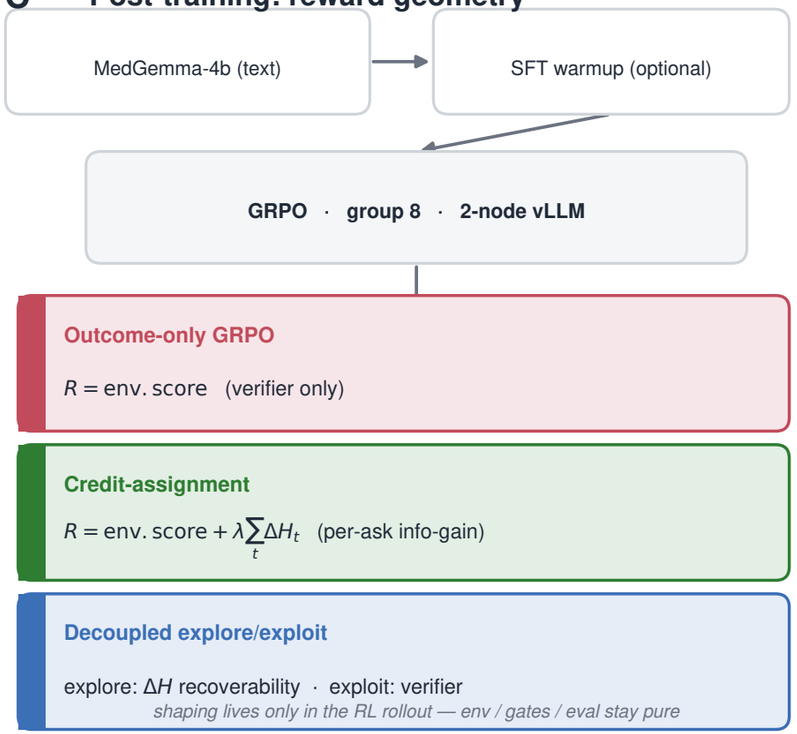
A Ask-to-commit loop



B Three ask-to-commit environments

same POMDP + exact adaptive oracle · diagnostics (highlighted) is the running example in D

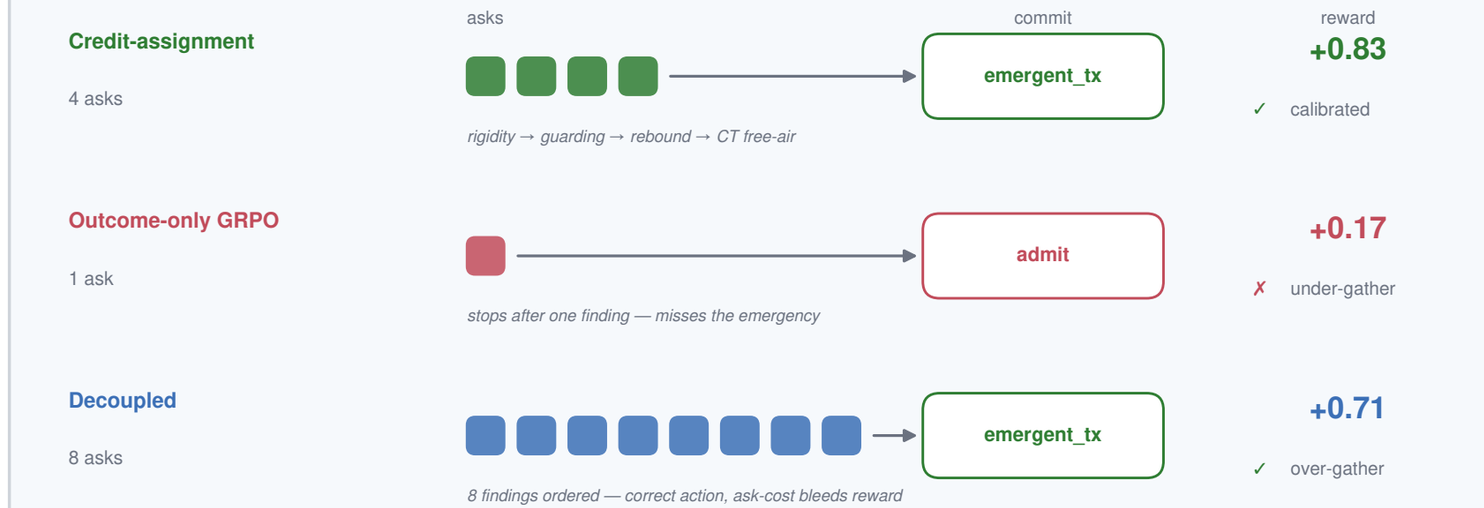
C Post-training: reward geometry



D Same patient, three procedures — one verifier

Patient dx-13: acute abdominal pain · latent = EMERGENT (hidden) · correct action = emergent_tx

held-out · reward shaping OFF · same env.score verifier for all three



Reward design changes the procedure — not just the score.